

Oliver Loo Chong Zheng

AI engineer and generalist. I put LLMs into products people actually use, and I ship them end to end.

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EDUCATION

- National University of Singapore — Bachelor of Computing (Honours), Computer Science**2021 – 2026
- Second Class Upper Honours (Distinction), CAP 4.02 / 5.00. Security specialisation. Degree conferred 31 August 2026.
- Stanford University — Undergraduate Summer Visitor, 8 credits transferred**Summer 2025
- GPA 4.00 / 4.00. Computer Graphics and Imaging (A+), Technology Entrepreneurship (A-).

EXPERIENCE

- DART Consulting and Training — Cyber Youth Instructor & Software Engineer Intern**Jan – Jun 2026
- Singapore
- Worked inside a cybersecurity training and consulting environment, deepening practical security knowledge beyond coursework into how security is operated and adopted across an organisation.
 - Developed and delivered technical material and supported training communications, translating security concepts into content non-specialists could learn from and act on.
- HTX, Home Team Science and Technology Agency — AI and Cybersecurity Intern**Jan – Jun 2025
- Singapore
- Built automated data collection and workflow orchestration systems for social media platforms.
 - Deployed and managed lightweight LLMs in virtualised Linux environments, integrating them into data collection and online interaction pipelines.
 - Applied advanced AI techniques including vision-language models, captcha solvers and decision-making algorithms.
- Singapore Armed Forces, 10C4I — Platoon Sergeant (Second Sergeant)**Aug 2019 – Jul 2020
- Led a team maintaining and troubleshooting a critical army network, configuring and managing network devices, and delivering the training programme that integrated new operators.

PROJECTS

- Levanta — AI personal training app, live on the App Store**2025 – 2026
- Solo-designed and shipped an iOS app in React Native and Expo with local-first storage in WatermelonDB and a Google Gemini coaching layer.
 - Natural-language workout logging parses free text into structured sets; every AI-proposed action renders as a confirmable widget so the model never mutates user data silently.
- funhop.app — Singapore night-planner, live**2026
- Built the full stack in Next.js and TypeScript on Supabase, plus a seven-stage Python ingestion pipeline that scrapes Telegram and publisher sites and uses an LLM to extract structured events.
 - Diagnosed and fixed silent data corruption affecting roughly 85% of the events table, traced to stripped query strings on signed image URLs and synthesised detail URLs returning 404.
- Langfun — Oui Chef! and Sushi Senpai — language-learning arcade games**2026
- Built one reusable game engine in Expo, Skia and Reanimated serving two titles, with all language content isolated to JSON and a theme file so a new title needs no engine changes.
 - Implemented a Leitner spaced-repetition scheduler and plausibility-constrained distractor selection so players learn the phrases rather than pattern-match the options.
- Bao Bae — retrieval-augmented couples therapy platform**2025
- Built a RAG system using FAISS vector search over a therapeutic knowledge base, combining clinical material with real-time conversation analysis to deliver grounded Emotionally Focused Therapy sessions.

SECURITY PRACTICE & CERTIFICATIONS

- Hands-on labs:** Offensive Security Proving Grounds, HackSmarter and Hack The Box. Enumeration, exploitation and privilege escalation across Linux and Windows targets.
- Malware analysis and detection:** completed the *Practical Malware Analysis* lab set (static triage, dynamic analysis in isolated VMs, disassembly), and built a PCAP analysis agent with an MCP server that surfaces suspicious flows into an LLM analyst loop.
- Certifications:** Google Cybersecurity Professional Certificate, AWS Cloud Essentials.

SKILLS

AI	LLM integration, RAG, FAISS, vector search, agents and tool use, MCP, prompt design, vision-language models, Gemini and OpenAI APIs, NLP
Languages	Python, TypeScript, JavaScript, Java, C, Rust, SQL, Bash
Building	React, React Native, Expo, Next.js, Node.js, Supabase, PostgreSQL, MongoDB, Git, Unix, Docker
Security	Penetration testing, malware analysis, network security, vulnerability assessment, Linux hardening